makes up the stuff of the universe (rocks and so forth) and what forces act on them.

Classical physics provides a straightforward picture of the world, but a number of crucial moves were made along the way to setting it up. Notice that we had to be very specific about what information we required to figure out what would happen to the rock: its position, its velocity, and the forces acting on it. We can think of those forces as being part of the outside world, and the important information about the rock itself as consisting of just its position and velocity. The acceleration of the rock at any moment in time, by contrast, is not something we need to specify; that's exactly what Newton's laws allow us to calculate from the position and the velocity.

Together, the position and velocity make up the *state* of any object in classical mechanics. If we have a system with multiple moving parts, the classical state of that entire system is just a list of the states of each of the individual parts. The air in a normal-sized room will have perhaps 10^{27} molecules of different types, and the state of that air would be a list of the position and velocity of every one of them. (Strictly speaking, physicists like to use the momentum of each particle, rather than its velocity, but as far as Newtonian mechanics is concerned the momentum is simply the particle's mass times its velocity.) The set of all possible states that a system could have is known as the *phase space* of the system.

The French mathematician Pierre-Simon Laplace pointed out a profound implication of the classical mechanics way of thinking. In principle, a vast intellect could know the state of literally every object in the universe, from which it could deduce everything that would happen in the future, as well as everything that had happened in the past. *Laplace's demon* is a thought experiment, not a realistic project for an ambitious computer scientist, but the implications of the thought experiment are profound. Newtonian mechanics describes a deterministic, clockwork universe.

The machinery of classical physics is so beautiful and compelling that it seems almost inescapable once you grasp it. Many great minds who came after Newton were convinced that the basic superstructure of physics had been solved, and future progress lay in figuring out exactly what realization of classical physics (which particles, which forces) was the right one to describe the universe as a whole. Even relativity, which was world-transforming in its own way, is a variety of classical mechanics rather than a replacement for it.